

CENG381 Stochastic Processes

Recurrence Relations - Answer Key 1

Q1 Solution

a) Characteristic equation:

Substitute $a(n) = r^n$ into $a(n) = 5a(n-1) - 6a(n-2)$:

$$r^n = 5r^{n-1} - 6r^{n-2}$$

Divide every term by r^{n-2} (valid for $r \neq 0$):

$$r^2 = 5r - 6$$

$$r^2 - 5r + 6 = 0 \quad \leftarrow \text{this is the characteristic equation}$$

b) Roots and general solution:

$$\text{Factor: } (r - 2)(r - 3) = 0$$

$$\text{Root } r_1 = 2, \quad \text{Root } r_2 = 3$$

Since $r_1 \neq r_2$, the general solution is:

$$a(n) = A \cdot 2^n + B \cdot 3^n$$

where A and B are constants to be found from initial conditions.

c) Apply initial conditions $a(0)=1$, $a(1)=4$:

From $a(0) = 1$:

$$A \cdot 2^0 + B \cdot 3^0 = 1$$

$$A + B = 1 \quad \dots (i)$$

From $a(1) = 4$:

$$A \cdot 2^1 + B \cdot 3^1 = 4$$

$$2A + 3B = 4 \quad \dots (ii)$$

From (i): $A = 1 - B$. Substitute into (ii):

$$2(1 - B) + 3B = 4$$

$$2 - 2B + 3B = 4$$

$$B = 2, \quad A = 1 - 2 = -1$$

Closed-form answer:

$$a(n) = -1 \cdot 2^n + 2 \cdot 3^n$$

Verification:

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$$a(0) = -1 + 2 = 1 \text{ (OK)}$$

$$a(1) = -2 + 6 = 4 \text{ (OK)}$$

$$a(2) = 5 \cdot 4 - 6 \cdot 1 = 20 - 6 = 14; \text{ formula: } -4 + 18 = 14 \text{ (OK)}$$

Q2 Solution

a) Setting up the correct recurrence:

At time n , species alive = (new offspring) + (species aged 1 that survive).

- Each species alive at $n-1$ produces 2 offspring $\rightarrow 2 \cdot a(n-1)$ new species
- Species aged exactly 1 at time n = species that were newborn at $n-1$
= $a(n-2)$ species (born at $n-2$, now aged 1, not yet 2 units old, so survive)
- Species aged 2 at time n die (they were born at $n-2$ and are now 2 units old)

Wait -- more carefully: a species born at time k is:

aged 0 at time k , aged 1 at time $k+1$, aged 2 at time $k+2$ (dies).

So at time n :

- Newborns (aged 0): $2 \cdot a(n-1)$ [each of the $a(n-1)$ species at $n-1$ reproduces]
- Aged 1 survivors: $a(n-1)$ - (those born at $n-2$ who are now aged 2 and die)

Actually the simplest way: total at n = newborns + survivors from $n-1$.

Survivors from $n-1$ = those at $n-1$ who are not yet 2 units old

= $a(n-1)$ minus those born at $n-3$ (they are now aged 2 and die)

As a result, for this particular model (all species age-2 die):

$$a(n) = 2 \cdot a(n-1) + 2 \cdot a(n-2)$$

where $2 \cdot a(n-1)$ = offspring of all alive at $n-1$,

and $2 \cdot a(n-2)$ accounts for the surviving generation.

(The exact model depends on assumptions; the given recurrence is accepted.)

With $a(0) = 100$, $a(1) = 2 \cdot 100 = 200$ (first generation of offspring).

b) Solve $a(n) = 2 \cdot a(n-1) + 2 \cdot a(n-2)$, $a(0)=100$, $a(1)=200$:

Characteristic equation (substitute $a(n) = r^n$, divide by $r^{(n-2)}$):

$$r^2 = 2r + 2$$

$$r^2 - 2r - 2 = 0$$

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Quadratic formula:

$$r = (2 \pm \sqrt{4 + 8}) / 2 = (2 \pm \sqrt{12}) / 2$$

$$r = 1 \pm \sqrt{3}$$

$$r_1 = 1 + \sqrt{3} \sim 2.732$$

$$r_2 = 1 - \sqrt{3} \sim -0.732$$

General solution:

$$a(n) = A(1+\sqrt{3})^n + B(1-\sqrt{3})^n$$

$$\text{Apply } a(0) = 100: A + B = 100 \quad \dots (i)$$

$$\text{Apply } a(1) = 200: A(1+\sqrt{3}) + B(1-\sqrt{3}) = 200$$

$$A + B + \sqrt{3}(A-B) = 200$$

$$100 + \sqrt{3}(A-B) = 200$$

$$A - B = 100/\sqrt{3} = 100\sqrt{3}/3 \quad \dots (ii)$$

From (i) and (ii):

$$A = 50 + 50\sqrt{3}/3 = 50(1 + \sqrt{3})/3 = 50(3+\sqrt{3})/3$$

$$B = 50 - 50\sqrt{3}/3 = 50(3-\sqrt{3})/3$$

Closed form:

$$a(n) = [50(3+\sqrt{3})/3](1+\sqrt{3})^n + [50(3-\sqrt{3})/3](1-\sqrt{3})^n$$

c) Find smallest n with $a(n) > 1,000,000$:

For large n the dominant term is $A(1+\sqrt{3})^n \sim 50(3+\sqrt{3})/3 * (2.732)^n$.

We need: $50(3+\sqrt{3})/3 * (2.732)^n > 1,000,000$

$$50 * (3 + 1.732)/3 * (2.732)^n > 1,000,000$$

$$50 * 1.577 * (2.732)^n > 1,000,000$$

$$78.87 * (2.732)^n > 1,000,000$$

$$(2.732)^n > 12,679$$

Taking logarithms:

$$n > \log(12,679) / \log(2.732) = 4.103 / 0.4365 \sim 9.40$$

So n = 10 is the first time $a(n) > 1,000,000$.

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Verify with exact values (using $a(n) = 2*a(n-1) + 2*a(n-2)$):

$a(0)=100$, $a(1)=200$, $a(2)=600$, $a(3)=1600$, $a(4)=4400$,
 $a(5)=12000$, $a(6)=32800$, $a(7)=89600$, $a(8)=244800$,
 $a(9)=669200$, $a(10)=1,828,000 > 1,000,000$ (OK)
($a(9) = 669,200 < 1,000,000$, so $n=10$ is correct.)

Q3 Solution

a) Homogeneous solution:

The associated homogeneous RR is $a(n) = 3*a(n-1)$.

This has characteristic root $r = 3$, so:

$a_h(n) = A * 3^n$ for any constant A .

b) Particular solution $a_p(n) = c*n + d$:

Substitute $a_p(n) = c*n + d$ into the full RR $a(n) = 3*a(n-1) + 2*n$:

Left side: $a_p(n) = c*n + d$

Right side: $3*a_p(n-1) + 2*n = 3*(c*(n-1) + d) + 2*n$
 $= 3*c*n - 3*c + 3*d + 2*n$
 $= (3c + 2)*n + (3d - 3c)$

Match coefficients of n :

$$c = 3c + 2 \Rightarrow -2c = 2 \Rightarrow c = -1$$

Match constant terms:

$$d = 3d - 3c = 3d - 3*(-1) = 3d + 3$$

$$d - 3d = 3 \Rightarrow -2d = 3 \Rightarrow d = -3/2$$

Particular solution: $a_p(n) = -n - 3/2$

c) General solution and applying $a(1) = 3$:

General solution:

$$a(n) = a_h(n) + a_p(n) = A * 3^n - n - 3/2$$

Apply the initial condition $a(1) = 3$:

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$$A \cdot 3^1 - 1 - 3/2 = 3$$

$$3A - 5/2 = 3$$

$$3A = 3 + 5/2 = 11/2$$

$$A = 11/6$$

Final closed-form answer:

$$a(n) = (11/6) \cdot 3^n - n - 3/2$$

Verification:

$$a(1) = (11/6) \cdot 3 - 1 - 3/2 = 11/2 - 1 - 3/2 = 11/2 - 5/2 = 6/2 = 3 \quad (\text{OK})$$

$$a(2) = 3 \cdot a(1) + 2 \cdot 2 = 9 + 4 = 13$$

$$\text{Formula: } (11/6) \cdot 9 - 2 - 3/2 = 33/2 - 2 - 3/2 = 33/2 - 7/2 = 26/2 = 13 \quad (\text{OK})$$